

Ship Codebase Audit Report

Date: 2026-03-11 **Scope:** `api/src/` , `web/src/` , `shared/src/` , `e2e/` **Branch:** `master` **Auditor:** Claude Code (automated + manual verification)

All 7 categories are baselined here. Individual category files with full methodology, EXPLAIN ANALYZE plans, and raw benchmark data are archived in `audit/01-*.md` through `audit/07-*.md` .

Category 1 — Type Safety

Metric	Baseline
Total any types	338
Total type assertions (as)	267
Total non-null assertions (!)	24
@ts-ignore / @ts-expect-error	0 / 1
Strict mode enabled	Yes (all packages)
Total violations	630
Top violation-dense files	UnifiedEditor.tsx (25 as), projects.ts (18 any), yjsConverter.ts (15 any), PropertiesPanel.tsx (13 as), y-protocols.d.ts (13 any)

Top Findings

#	Finding	Severity
1	Unsafe document subtype casting in UnifiedEditor.tsx and PropertiesPanel.tsx — no discriminated union or type guards; runtime crashes if document shape diverges	High
2	any throughout API route handlers for DB rows (projects.ts, weeks.ts) — schema changes invisible to compiler	High
3	Yjs conversion pipeline fully untyped (yjsConverter.ts) — malformed CRDT data reaches the editor silently	Medium
4	web/tsconfig.json missing noUncheckedIndexedAccess and noImplicitReturns — weaker type config than the backend	Medium

Improvement Target

- Eliminate 25% of violations (~158) in Phase 2
- Add DB row types in `projects.ts` (~18 any removed)
- Introduce discriminated union in `UnifiedEditor.tsx` (~25 as removed)
- Type TipTap JSON schema in `yjsConverter.ts` (~15 any removed)
- Align `web/tsconfig.json` to extend root config

Category 2 — Bundle Size

Metric	Baseline
Total production JS (raw / gzip)	2,197 KB / ~620 KB
Largest chunk	index.js — 2,073 KB raw / 589 KB gzip (94.4% of total JS)
Number of chunks	261 (1 main + 14 tab components + 246 SVG icons)
Top 3 heaviest dependencies	@tiptap/core + extensions (3.3 MB dist), emoji-picker-react (2.3 MB dist), yjs + y-socket + y-indexeddb (2.2 MB)
Unused dependencies in prod	@tanstack/react-query-devtools — in dependencies (not devDependencies), shipped unconditionally to all users

Top Findings

#	Finding	Severity
1	94% of JS in one monolithic chunk — Vite warns at 500 KB; zero page-level code splitting	High
2	@tanstack/react-query-devtools ships to production — ~256 KB of dev tooling downloaded on every page load	High
3	emoji-picker-react (2.3 MB dist) loads on every page — statically imported, only used in one popover	Medium–High
4	All 15+ page routes are static imports in main.tsx — TipTap, Yjs, and all providers initialize on first load regardless of page	Medium–High

Improvement Target

- Target 15% reduction in total bundle OR 20% reduction in initial load
- Move react-query-devtools to devDependencies + NODE_ENV guard (immediate ~256 KB saving)
- Convert page imports in main.tsx to React.lazy() (largest potential gain)
- Lazy-load EmojiPicker.tsx on popover open

Category 3 — API Response Time

P95 column uses p97.5 (closest available autocannon bucket). All values at c=10 (single-user equivalent).

Endpoint	P50	P95 (p97.5)	P99
GET /api/issues	18ms	32ms	36ms
GET /api/weeks/my-week	7ms	28ms	31ms
GET /api/projects	5ms	18ms	21ms

GET /api/documents?type=wiki	23ms	32ms	33ms
GET /api/dashboard/my-work	6ms	20ms	22ms

At c=50 (peak load), worst performers:

Endpoint	P99 (c=50)
GET /api/documents?type=wiki	142ms (4.3× degradation from c=10)
GET /api/issues	120ms (3.3× degradation)

Top Findings

#	Finding	Severity
1	GET /api/issues fetches full d.content for all 150 issues — list UI only needs title/state/priority/assignee	High
2	GET /api/documents degrades 4.3× from c=10 to c=50 — driven by pg-pool contention (default 10 connections)	High
3	Auth middleware runs 3 DB queries on every request — at c=50 this is 150 concurrent session writes	Medium–High
4	Sequential DB queries in dashboard and projects routes — could run in parallel via Promise.all	Medium

Improvement Target

- 20% reduction in p99 on at least 2 endpoints
- Remove d.content from issues list SELECT (GET /api/issues target: ≤96ms p99 at c=50)
- Increase pg-pool max from 10 to 25 (GET /api/documents target: ≤114ms p99 at c=50)
- Throttle UPDATE sessions SET last_activity to once per 60s per session

Category 4 — Database Query Efficiency

User Flow	Total DB Queries	Slowest Query	N+1 Detected?
Load main page	25	1.41ms	No
View a document	4	0.08ms	No
List issues	5	1.74ms	No
Load sprint board	16	1.38ms	No
Search content	9	1.12ms	No

Main page query breakdown: 15 of 25 queries (60%) are auth middleware overhead (3 queries × 5 parallel requests). Route logic accounts for only 10 queries.

Top Findings

#	Finding	Severity
1	Auth middleware: 3 DB queries per request — SELECT session+user, SELECT workspace_membership, UPDATE last_activity (unconditional)	High
2	GET /api/issues selects d.content for all 150 issues; list UI discards it	High
3	GET /api/documents?type=wiki performs full sequential scan (556 rows, 322 discarded) — composite index idx_documents_active is unusable because route query omits deleted_at IS NULL	Medium–High
4	idx_documents_active has dead partial condition (deleted_at IS NULL — 0 rows in table have a deleted_at value)	Medium
5	N+1 in GET /api/weeks/:id/scope-changes — one query per removed issue inside a loop; should be batched	Medium
6	No functional index on (properties->>'assignee_id') — dashboard query discards 148/150 rows after a JSONB text comparison	Medium

Improvement Target

- 20% reduction in query count on at least one flow, or 50% improvement on slowest query
- Throttle session UPDATE → removes 5 queries from main page load (25 → 20)
- Merge 2 auth SELECTs into single JOIN → removes 5 more (20 → 15, a 40% reduction)
- Add AND deleted_at IS NULL to wiki query to enable composite index
- Add functional index: CREATE INDEX idx_documents_assignee ON documents ((properties->>'assignee_id')) WHERE document_type = 'issue'
- Batch the scope-changes N+1 with WHERE id = ANY(\$1::uuid[])

Category 5 — Test Coverage

Metric	Baseline
Total tests written	1,479 (451 API + 162 web + 866 E2E)
Tests currently passing	451 (API unit only — the only suite that runs via pnpm test)
Suite runtime	12.7s (API unit)
API statement coverage	40.34%
API branch coverage	33.44%
Web coverage	0% (test suite broken — ESM/CJS dependency error; 0 of 162 web tests execute)
Critical flows with zero coverage	CAIA/OAuth government login, WebSocket disconnect recovery, concurrent two-user edit conflict

Top Findings

#	Finding	Severity
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1	All 16 web unit test files broken (ERR_REQUIRE_ESM) — 162 written tests, 0 running; frontend has no passing unit coverage	High
2	CAIA/OAuth government login path has zero test coverage — the only real auth path for Treasury users is completely untested	High
3	API branch coverage 33% — two-thirds of conditional logic paths (error branches, auth edge cases) run untested in production	Medium–High
4	pnpm test only runs API unit tests — a passing pnpm test result hides 16 broken web files and the entire E2E suite	Medium
5	No multi-browser E2E test for real-time collaboration — a regression in the WebSocket/Yjs layer would not be caught	Medium

Improvement Target

- Fix web unit test ESM/CJS breakage (restores 162 tests)
- Add CAIA auth unit tests covering: success, new user provisioning, .gov/.mil rejection, invalid CSRF state, expired token
- Add two-browser Playwright E2E test for collaborative editing

Category 6 — Runtime Error Handling

Metric	Baseline
Console errors during normal usage	1 — 401 Unauthorized from /api/auth/me on initial load before auth redirect
Unhandled promise rejections (server)	No global handler — Node.js 15+ terminates the process on unhandled rejection with only a stderr log
Network disconnect recovery	Partial — editor buffers changes in IndexedDB; non-editor pages show stale data silently
Root-level error boundary	None — main.tsx has no ErrorBoundary; 8 providers are unprotected
Silent failures identified	6

Silent Failures

#	Failure	Severity
1	No root-level ErrorBoundary in main.tsx — any provider crash produces a blank white screen; App.tsx:542 only covers page content	High
2	useRealtimeEvents reconnects every 3s on any close including 429 — no backoff, no 429 awareness; client hammers server indefinitely	High
3	Editor WebSocket has no 429 handling — on rate-limit, editor shows Offline indefinitely with no explanation	Medium
4	Title <textarea> has no maxLength — saves silently fail (400) above 255 chars; title reverts on next sync with no message	Medium

5	ROLLBACK errors silently swallowed in <code>documents.ts</code> and <code>issues.ts</code> — <code>.catch(() => {})</code> discards rollback failures	Medium
6	No <code>process.on('unhandledRejection')</code> handler — any uncaught async throw terminates the API process with only a <code>stderr</code> log	Medium

Improvement Target

- Fix 3 error handling gaps with user-facing impact
- Add exponential backoff (3s → 6s → 12s → 24s → 60s max) with 429 awareness to `useRealtimeEvents`
- Show "Connection blocked — changes saved locally" after 3 consecutive 1006 closes in `Editor.tsx`
- Add `maxLength={255}` + character counter to the document title textarea

Category 7 — Accessibility

Metric	Baseline
Lighthouse accessibility score (per page)	My week main /my-week — 96/100 Docs main view /docs — 100/100 Docs selected wiki /docs/:id — 100/100 Selected Program view /docs/:id — 94/100 Selected Programs – issues view /docs/:id/issues — 95/100 Selected Programs – projects view /docs/:id/projects — 91/100 Selected Programs – weeks view /docs/:id/weeks — 91/100 Selected issues view /docs/:id — 95/100 Selected Projects view /docs/:id — 95/100 Selected Projects details view /docs/:id/details — 92/100 Selected Projects week view /docs/:id/weeks — 91/100 Selected Projects retro view /docs/:id/retro — 92/100 Program main view /programs — 100/100 Projects main view /projects — 96/100 Team allocation /team/allocation — 96/100 Team directory /team/directory — 100/100 Selected team member /team/:id — 100/100 Team status /team/status — 96/100 Team reviews /team/reviews — 96/100 Team org-chart /team/org-chart — 100/100 Settings /settings — 95/100 Settings invites /settings?tab=invites — 94/100 Settings API tokens /settings?tab=tokens — 93/100 Settings audit logs /settings?tab=audit — 100/100 Settings conversions /settings/conversions — 100/100
Total Critical / Serious violations	3 Critical, 10 Serious — see below

Keyboard navigation completeness	Partial — 5 broken/missing flows
Color contrast failures	6 failures
Missing ARIA labels or roles	8 locations

Critical Violations

#	Page / Flow	Violation
1	/documents/:id — any open document	aria-expanded="false" on ProseMirror root role="textbox" — attribute not allowed; AT may skip or misinterpret the entire editor
2	/documents/:id — empty title field	Title placeholder contrast ~1.6:1 (needs 4.5:1): #8a8a8a at 30% opacity → effective ~`#2a2a2a`
3	All pages — provider crash	No root-level <ErrorBoundary> in main.tsx; 8 providers unprotected; any exception → blank white screen

Serious Violations

#	Page / Flow	Violation
4	/documents/:id — backlog picker	BacklogPickerModal.tsx:244 search input missing aria-label
5	/documents/:id — properties sidebar, association chips	MultiAssociationChips.tsx:172 search input missing aria-label
6	Any page — Cmd+K / Ctrl+K	CommandPalette.tsx:254 search input missing aria-label
7	/documents/:id — document header	EmojiPicker.tsx:56 trigger button missing aria-label
8	/programs, /docs/:id — merge program dialog	MergeProgramDialog search placeholder contrast ~3.2:1 (needs 4.5:1)
9	/my-week, /docs/:id — any tabbed view	TabBar no ArrowLeft/ArrowRight navigation (WCAG 2.1 SC 2.1.1 violation)
10	/documents/:id — properties sidebar	PropertyRow.tsx:15 labels not associated with inputs via htmlFor/id
11	/documents/:id — properties sidebar	PropertyRow.tsx:21 required field indicator missing aria-describedby/aria-required
12	/projects, /docs/:id/issues, approval flow	Decorative SVG icons in BulkActionBar, ApprovalButton, CommandPalette missing aria-hidden="true"
13	/my-week — weekly plan editor	text-muted/50 line numbers: ~3.2:1 contrast (needs 4.5:1)

Color Contrast Failures

Location	Effective Ratio	Required	Result
/documents/:id title placeholder (#8a8a8a @ 30%)	~1.6:1	4.5:1	FAIL

Merge dialog search placeholder (#8a8a8a @ 50%)	~3.2:1	4.5:1	FAIL
/my-week line numbers (text-muted/50)	~3.2:1	4.5:1	FAIL
/settings, /programs dash text (text-muted/50)	~3.2:1	4.5:1	FAIL
/documents/:id body placeholder (#525252)	~3.2:1	4.5:1	FAIL
Disabled buttons (opacity-50 on bg-accent) — all pages	~1.8:1	4.5:1	FAIL

Improvement Target

Goal A — Raise /docs/:id/projects from 91/100 to 100/100:

- Add aria-hidden="true" to decorative SVGs in BulkActionBar
- Add focus-visible:ring to SelectableList rows
- Replace text-muted/50 with a full-opacity token on content text in WorkspaceSettings

Goal B — Fix all critical & serious violations on 3 most important pages:

Page	Fixes
/documents/:id	Remove aria-expanded from ProseMirror textbox root; fix title placeholder contrast; add aria-label to EmojiPicker; associate PropertyRow labels; add aria-label to MultiAssociationChips
/my-week	Fix line number contrast (text-muted/50 → solid token); implement TabBar arrow key navigation
/projects	Add aria-hidden="true" to BulkActionBar/ApprovalButton SVGs; add focus ring to SelectableList rows

Cross-Category Summary

Category	Key Risk	Priority
Type Safety	630 violations; unsafe document casting can crash the editor silently	Medium
Bundle Size	2.1 MB monolithic chunk; devtools shipped to production users	High
API Response Time	GET /api/documents degrades 4.3x under load; issues payload oversized	High
DB Query Efficiency	60% of page-load queries are auth overhead; sequential scan on wiki list	High
Test Coverage	Web tests fully broken (0%); CAIA auth path zero coverage	High
Runtime Error Handling	No root ErrorBoundary; WS reconnect storm on 429; silent title revert	High
Accessibility	Critical ARIA violation on editor root (every document); 6 contrast failures	High

Highest-Impact Phase 2 Fixes (Quick Wins)

1. Move `react-query-devtools` to `devDependencies` — 1 line change, saves ~256 KB for all users
2. Add root `<ErrorBoundary>` in `main.tsx` — 5 lines, eliminates blank-screen crash for all users
3. Add `aria-label` to `CommandPalette`, `BacklogPickerModal`, `MultiAssociationChips` inputs — 3 attribute additions
4. Throttle session `UPDATE` to once per 60s — removes 5 queries from every page load (20% reduction)
5. Remove `d.content` from issues list `SELECT` — 1 column removed, directly cuts issues payload
6. Fix `useRealtimeEvents` `backoff` — prevents reconnect storm from making rate-limits permanent